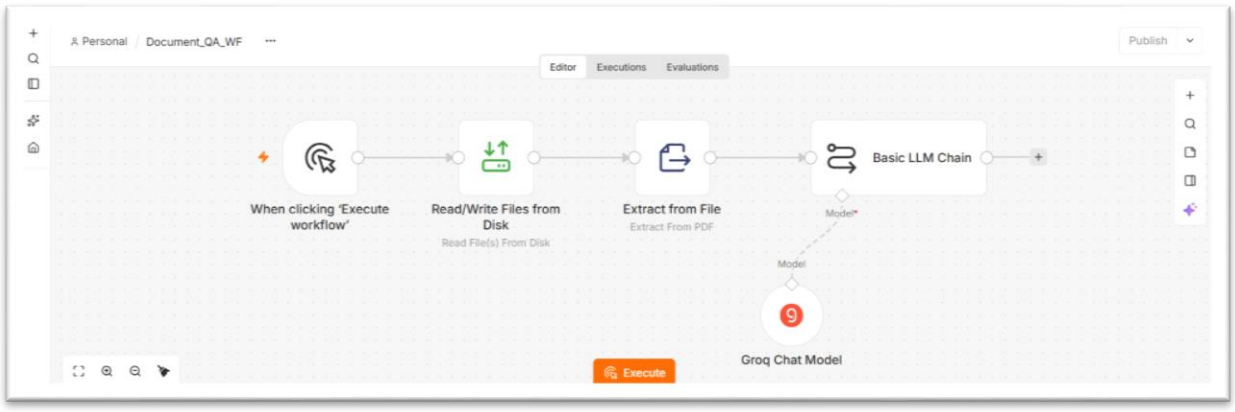
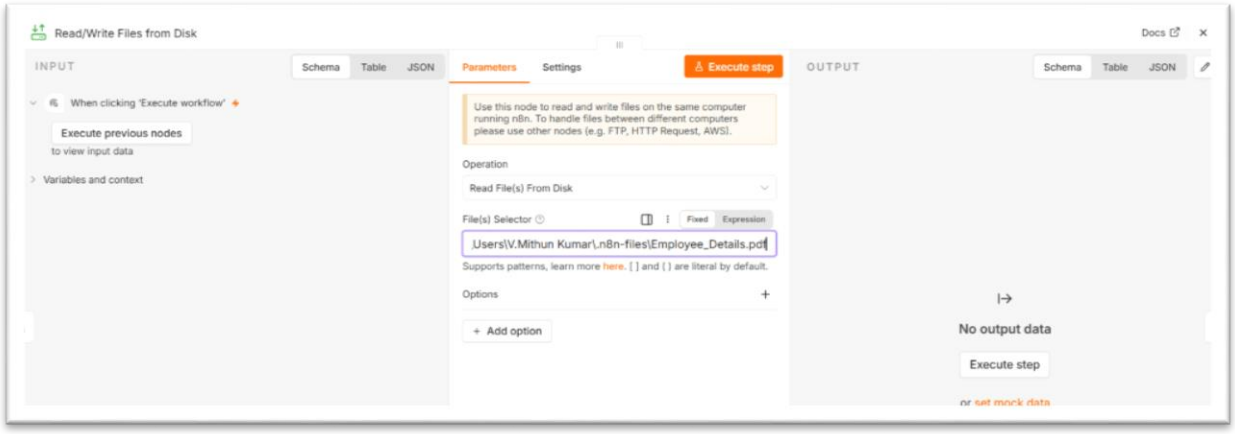
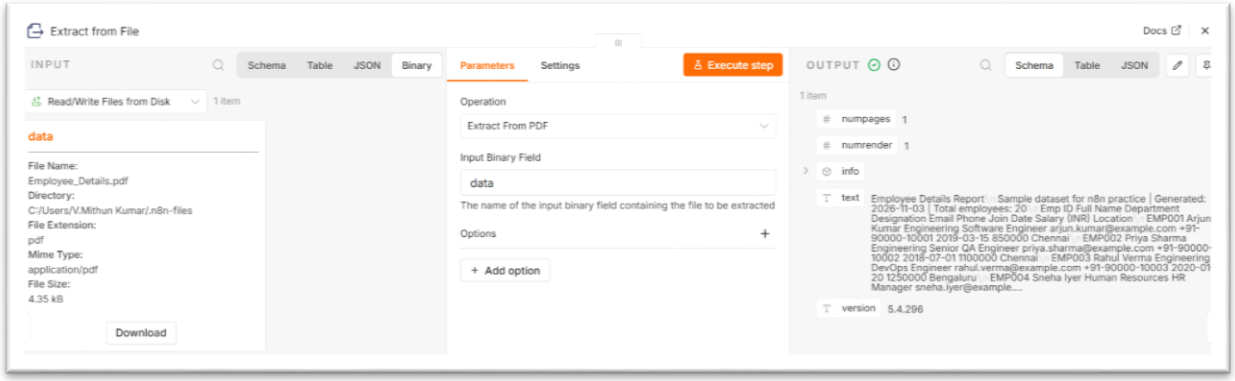
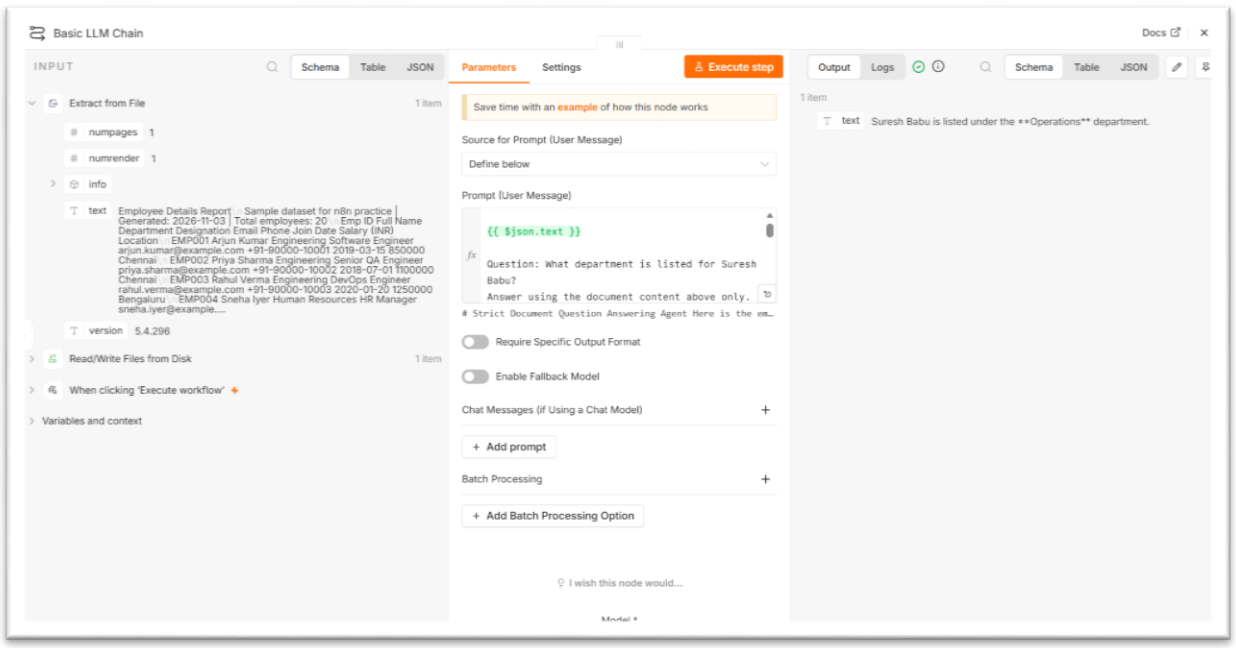
Document Question Answering (Document QA) Agent

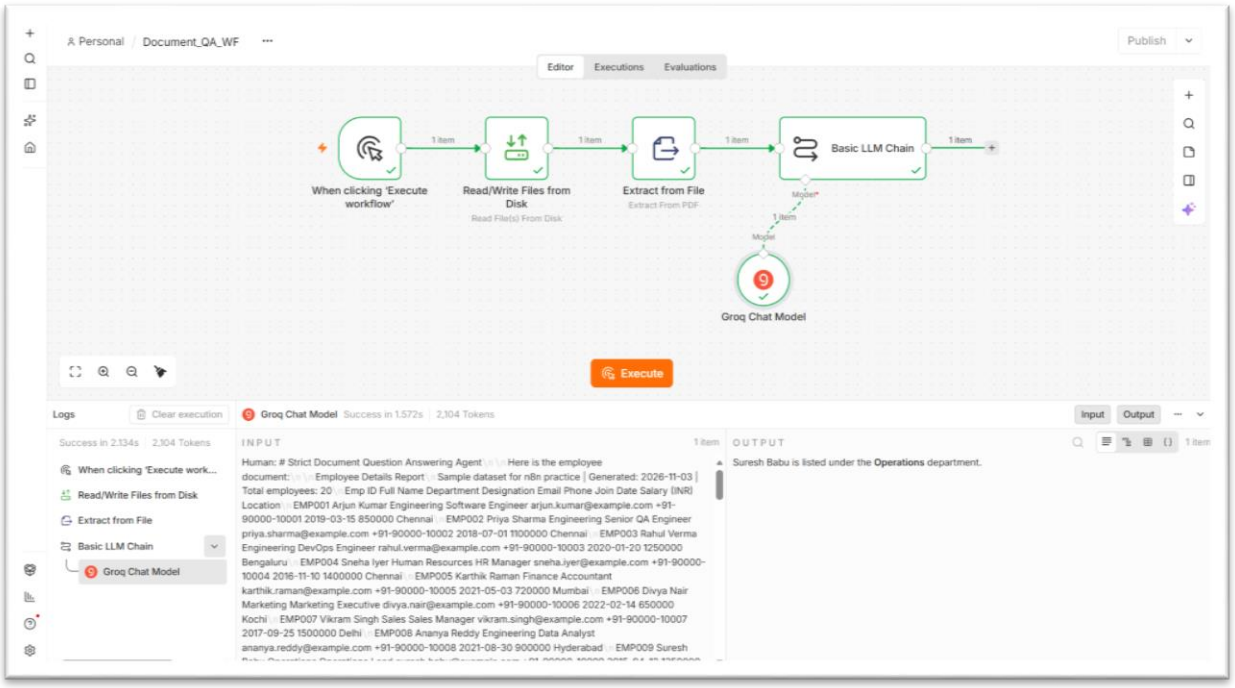
Phase 1: Document QA with Predefined PDF Document and User Question








Basic LLM Chain
Docs
INPUT
Schema
Table
JSON
Parameters
Settings
Execute step
Output
Logs
Extract from File
1 item
numpages 1
numrender 1
info
text Employee Details Report | Sample dataset for n8n practice | Generated: 2026-11-03 | Total employees: 20 | Emp ID Full Name Department Designation Email Phone Join Date Salary (INR) Location | EMP001 Arjun Kumar Engineering Software Engineer arjun.kumar@example.com +91-90000-10001 2019-03-15 850000 Chennai | EMP002 Priya Sharma Engineering Senior QA Engineer priya.sharma@example.com +91-90000-10002 2018-07-01 1100000 Chennai | EMP003 Rahul Verma Engineering DevOps Engineer rahul.verma@example.com +91-90000-10003 2020-01-20 1250000 Bengaluru | EMP004 Sneha Iyer Human Resources HR Manager sneha.iyer@example....
version 5.4.296
Read/Write Files from Disk
1 item
When clicking 'Execute workflow'
Variables and context
Save time with an example of how this node works
Source for Prompt (User Message)
Define below
Prompt (User Message)
{{ $json.text }}
Question: What department is listed for Suresh Babu?
Answer using the document content above only.
Strict Document Question Answering Agent Here is the em...
Require Specific Output Format
Enable Fallback Model
Chat Messages (if Using a Chat Model)
Add prompt
Batch Processing
Add Batch Processing Option
I wish this node would...
Model *
1 item
text Suresh Babu is listed under the **Operations** department.


Personal / Document_QA_WF
Publish
Editor
Executions
Evaluations
When clicking 'Execute workflow'
Read/Write Files from Disk
Read File(s) From Disk
Extract from File
Extract From PDF
Basic LLM Chain
Model*
Groq Chat Model
Execute
Logs
Clear execution
Success in 2.134s | 2,304 Tokens
When clicking 'Execute work...
Read/Write Files from Disk
Extract from File
Basic LLM Chain
Groq Chat Model
Groq Chat Model Success in 1.572s | 2,104 Tokens
Input
Output
INPUT
1 item
Human: # Strict Document Question Answering Agent Here is the employee document: Employee Details Report | Sample dataset for n8n practice | Generated: 2026-11-03 | Total employees: 20 | Emp ID Full Name Department Designation Email Phone Join Date Salary (INR) Location | EMP001 Arjun Kumar Engineering Software Engineer arjun.kumar@example.com +91-90000-10001 2019-03-15 850000 Chennai | EMP002 Priya Sharma Engineering Senior QA Engineer priya.sharma@example.com +91-90000-10002 2018-07-01 1100000 Chennai | EMP003 Rahul Verma Engineering DevOps Engineer rahul.verma@example.com +91-90000-10003 2020-01-20 1250000 Bengaluru | EMP004 Sneha Iyer Human Resources HR Manager sneha.iyer@example.com +91-90000-10004 2016-11-10 1400000 Chennai | EMP005 Karthik Raman Finance Accountant karthik.raman@example.com +91-90000-10005 2021-05-03 720000 Mumbai | EMP006 Divya Nair Marketing Marketing Executive divya.nair@example.com +91-90000-10006 2022-02-14 650000 Kochi | EMP007 Vikram Singh Sales Sales Manager vikram.singh@example.com +91-90000-10007 2017-09-25 1500000 Delhi | EMP008 Ananya Reddy Engineering Data Analyst ananya.reddy@example.com +91-90000-10008 2021-08-30 900000 Hyderabad | EMP009 Suresh
OUTPUT
1 item
Suresh Babu is listed under the Operations department.

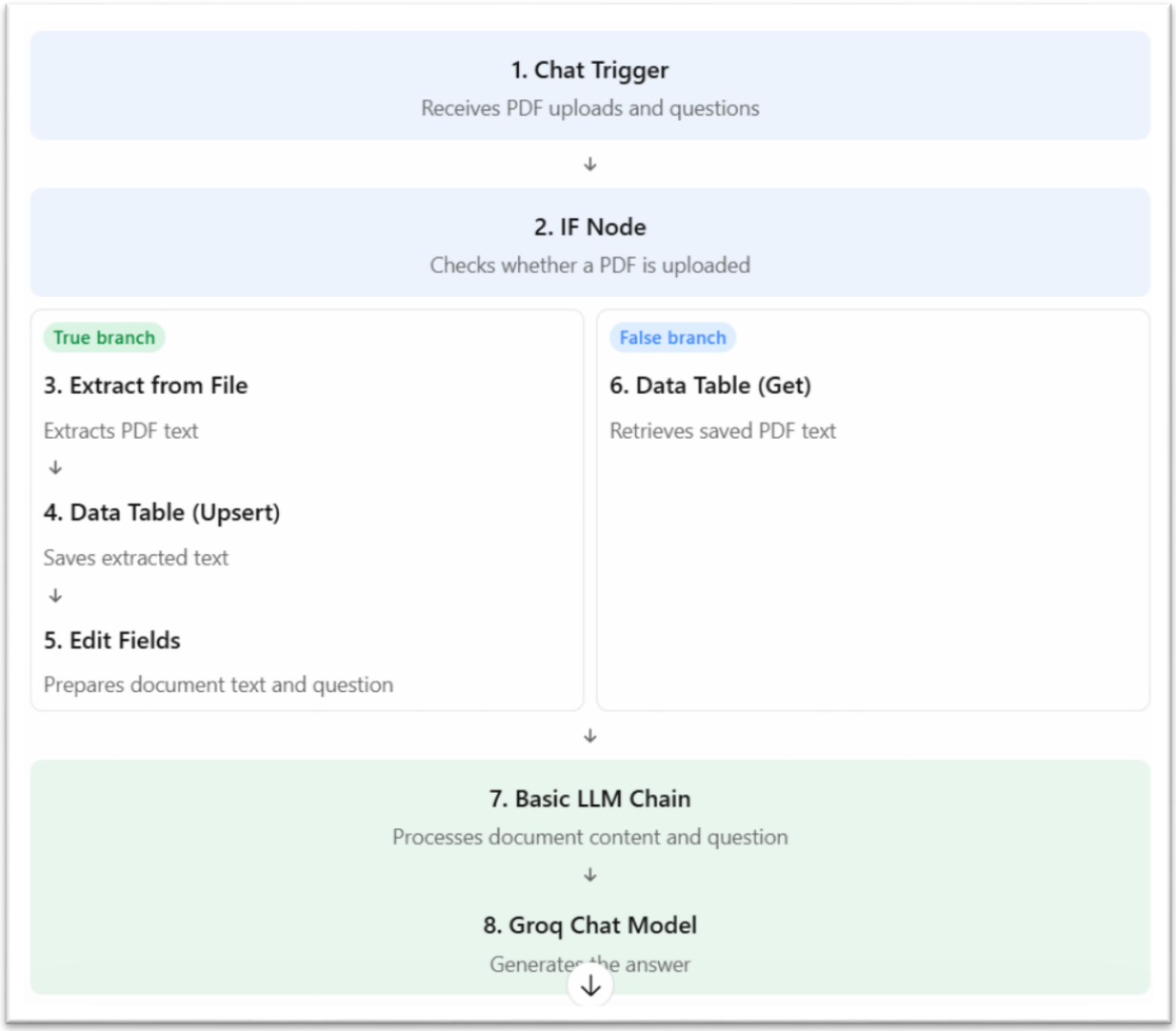
Phase 2: Document QA with Runtime PDF Upload, Dynamic User Questions and Session-Based Retrieval

Objective: Build a Document QA Agent that allows users to upload PDF documents at runtime and ask multiple questions dynamically through chat.

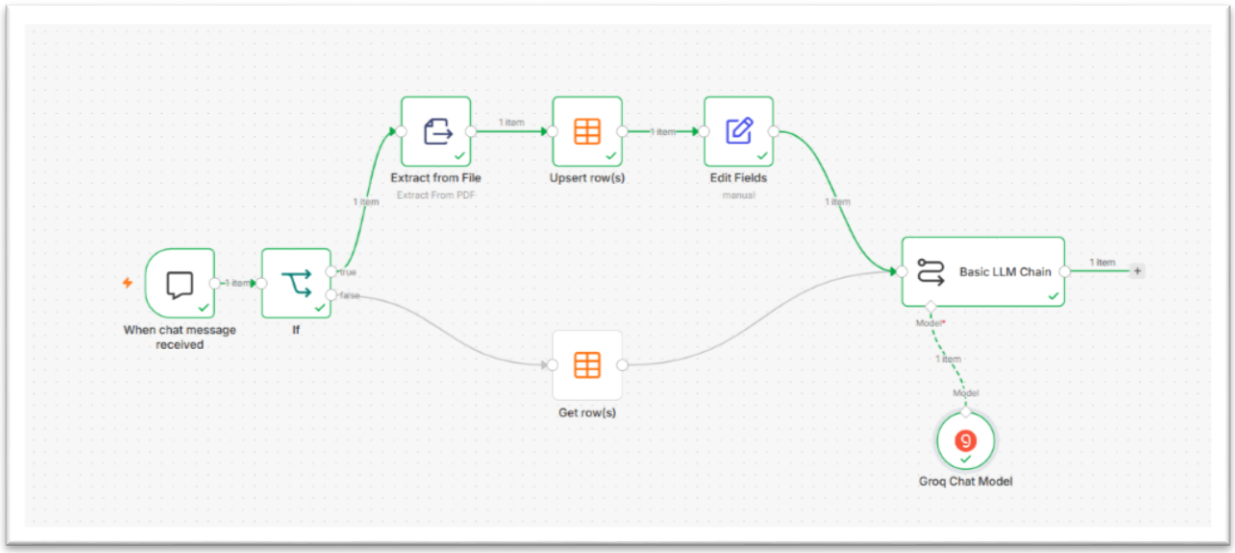
Key functionalities:

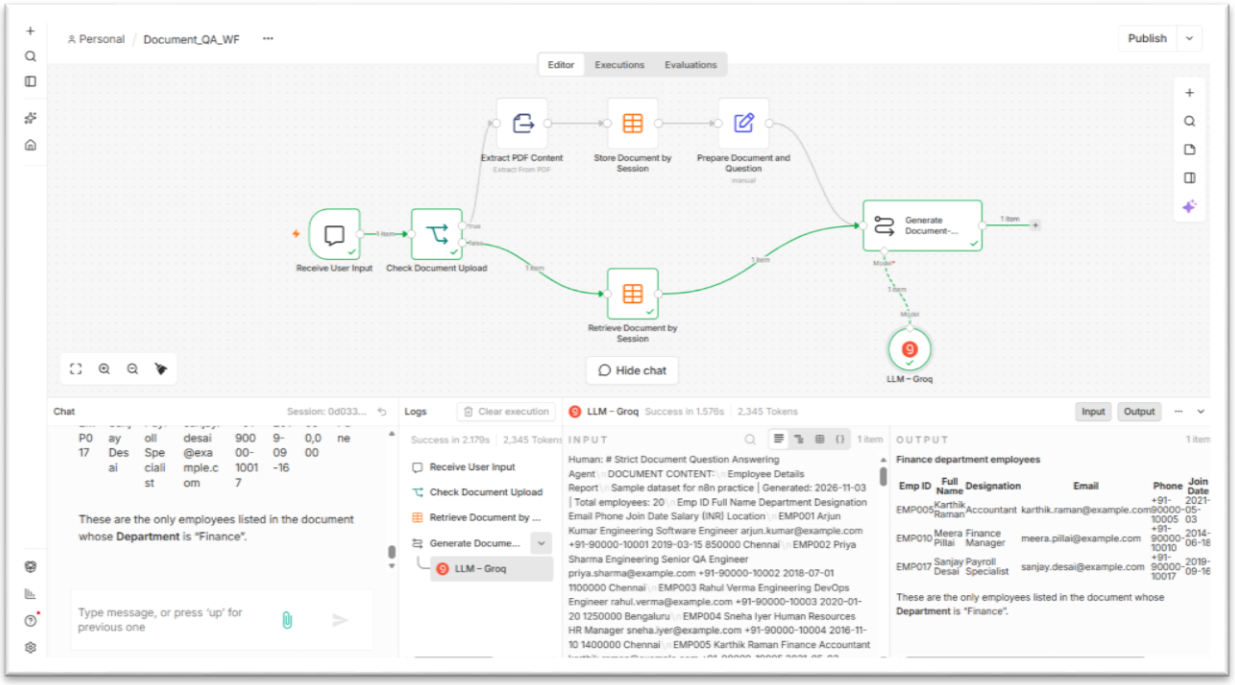
1. Runtime PDF Upload: Users can upload PDF documents directly through the chat interface.
2. PDF Text Extraction: Extracts readable content using the Extract from File node.
3. Session-Based Storage: Stores extracted document content in an n8n Data Table using sessionId.
4. Dynamic Question Input: Accepts user questions through Chat Trigger instead of hardcoding questions in the prompt.
5. Follow-up Questions: Retrieves the previously uploaded document using Data Table (Get), allowing multiple questions within the same chat session.
6. AI-Based Answer Generation: Uses Basic LLM Chain and Groq Chat Model to generate answers based strictly on the uploaded document.
7. Missing Information Handling: Returns an appropriate message when the requested information is unavailable in the uploaded PDF.

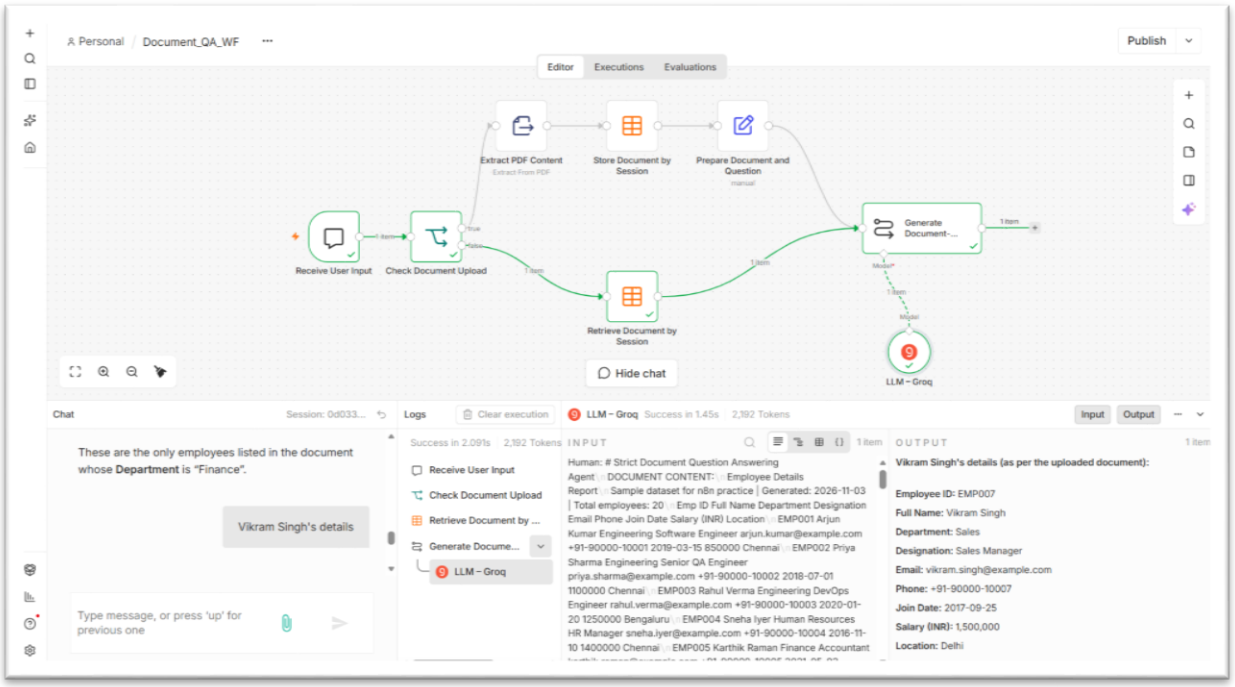
Overall Workflow:

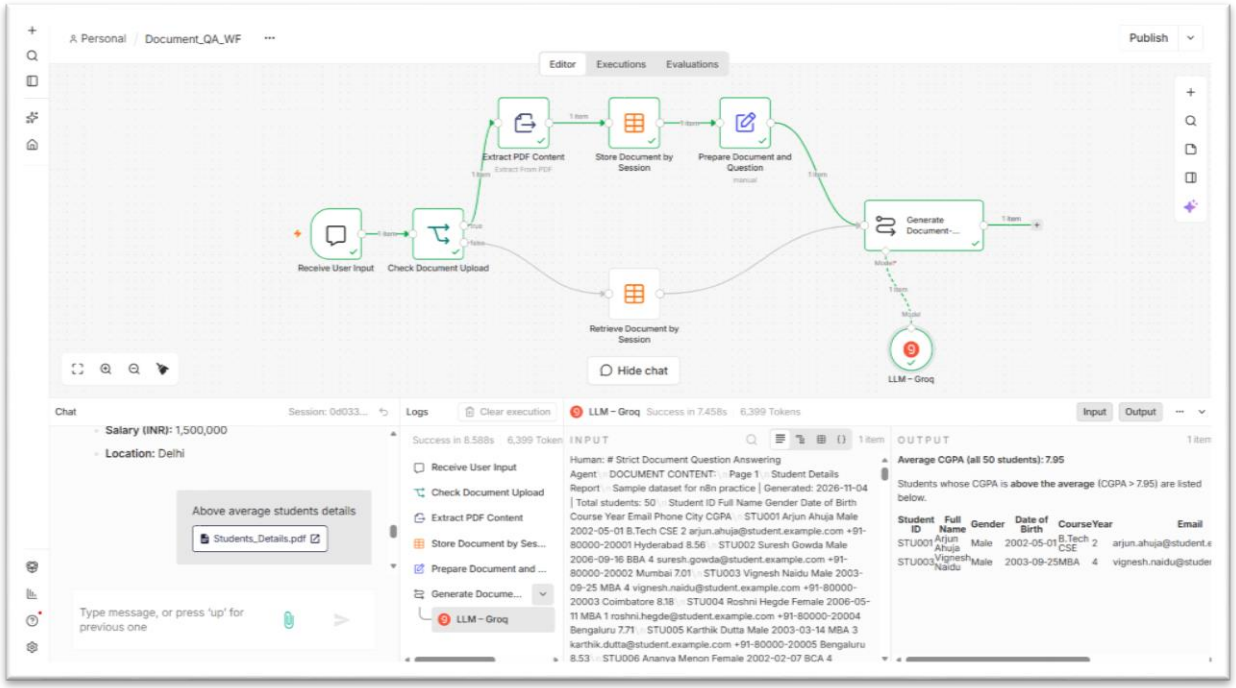


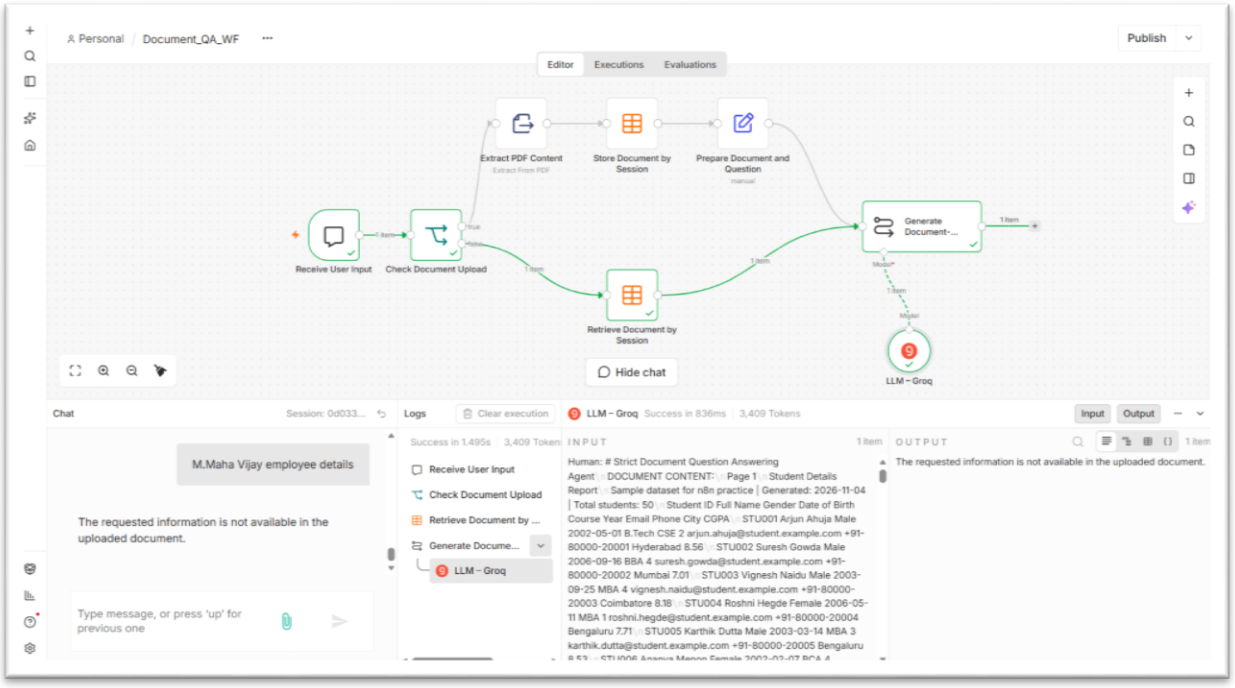
n8n Workflow:

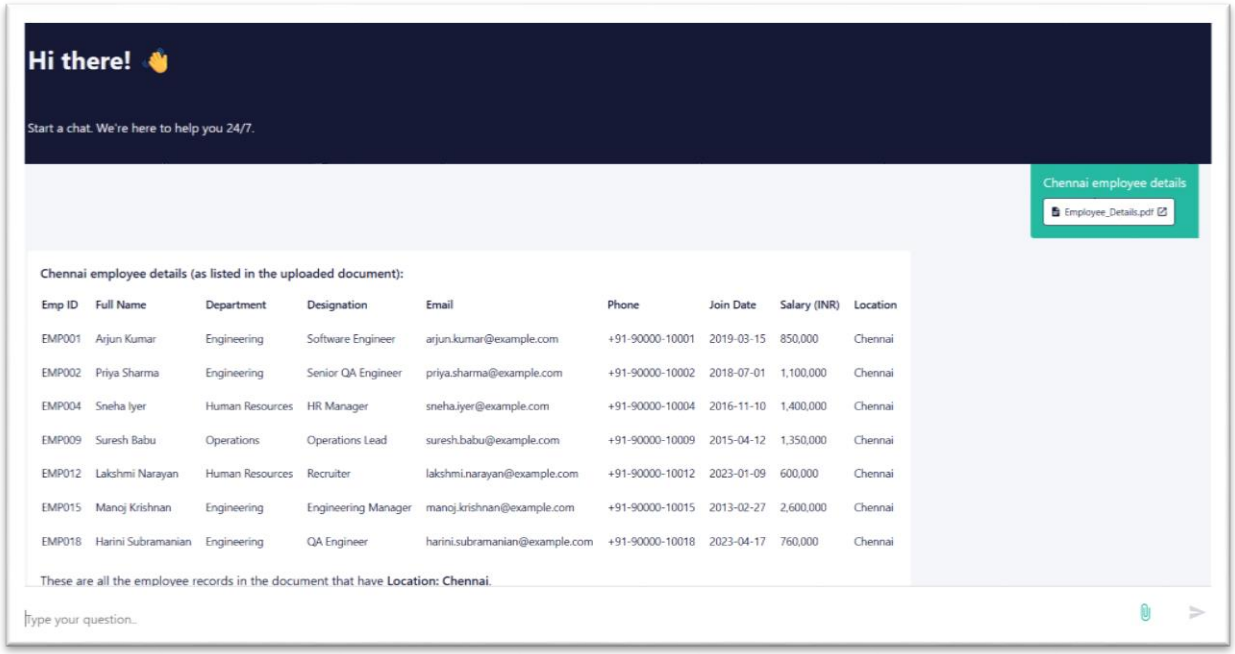


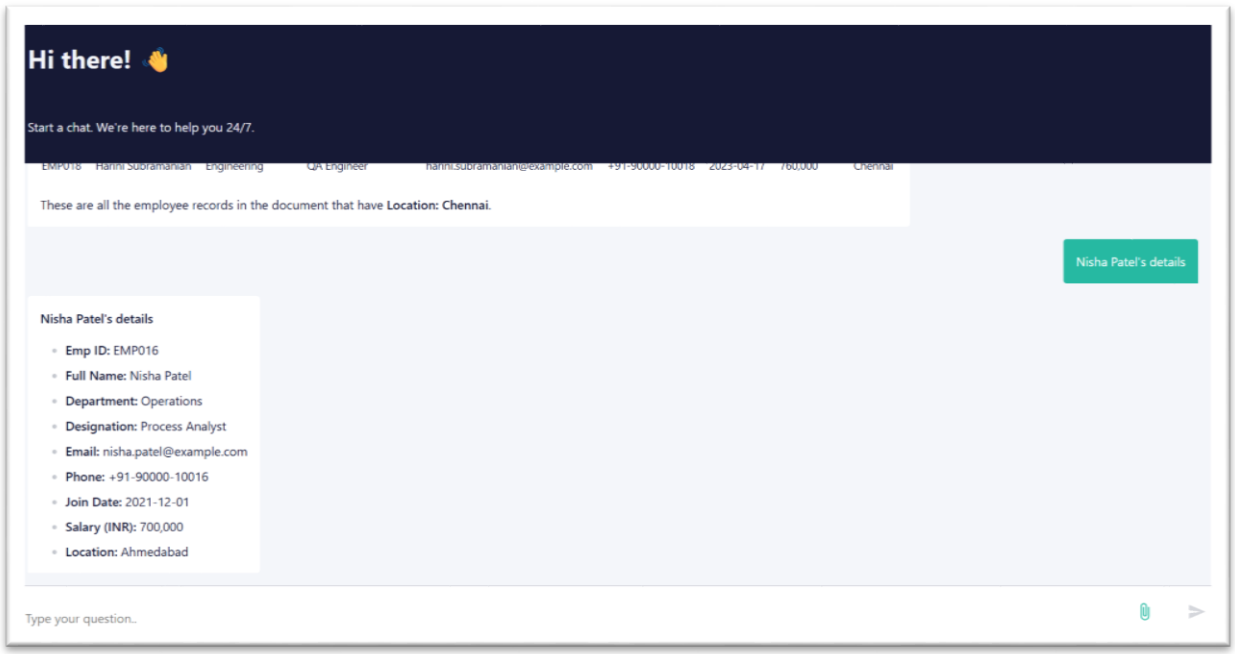

Personal / Document_QA_WF
Publish
Editor
Executions
Evaluations
Extract PDF Content
Store Document by Session
Prepare Document and Question
Receive User Input
Check Document Upload
Retrieve Document by Session
Generate Document-...
LLM – Groq
Hide chat
Chat
Logs
Clear execution
LLM – Groq Success in 1.576s 2,345 Tokens
Input
Output
Receive User Input
Check Document Upload
Retrieve Document by ...
Generate Docume...
LLM – Groq
INPUT
OUTPUT
Finance department employees
Emp ID
Full Name
Designation
Email
Phone
Join Date
EMP005 Karthik Raman Accountant karthik.raman@example.com +91-90000-10005 2021-05-03
EMP010 Meera Pillai Finance Manager meera.pillai@example.com +91-90000-10010 2014-06-18
EMP017 Sanjay Desai Payroll Specialist sanjay.desai@example.com +91-90000-10017 2019-09-16
These are the only employees listed in the document whose Department is "Finance".
These are the only employees listed in the document whose Department is "Finance".
Type message, or press 'up' for previous one


Personal / Document_QA_WF
Publish
Editor
Executions
Evaluations
Extract PDF Content
Store Document by Session
Prepare Document and Question
Receive User Input
Check Document Upload
Retrieve Document by Session
Generate Document-...
LLM – Groq
Hide chat
Chat
Logs
Clear execution
LLM – Groq Success in 1.45s 2,192 Tokens
Input
Output
Receive User Input
Check Document Upload
Retrieve Document by ...
Generate Docume...
LLM – Groq
INPUT
OUTPUT
Vikram Singh's details (as per the uploaded document):
Employee ID: EMP007
Full Name: Vikram Singh
Department: Sales
Designation: Sales Manager
Email: vikram.singh@example.com
Phone: +91-90000-10007
Join Date: 2017-09-25
Salary (INR): 1,500,000
Location: Delhi
These are the only employees listed in the document whose Department is "Finance".
Vikram Singh's details
Type message, or press 'up' for previous one

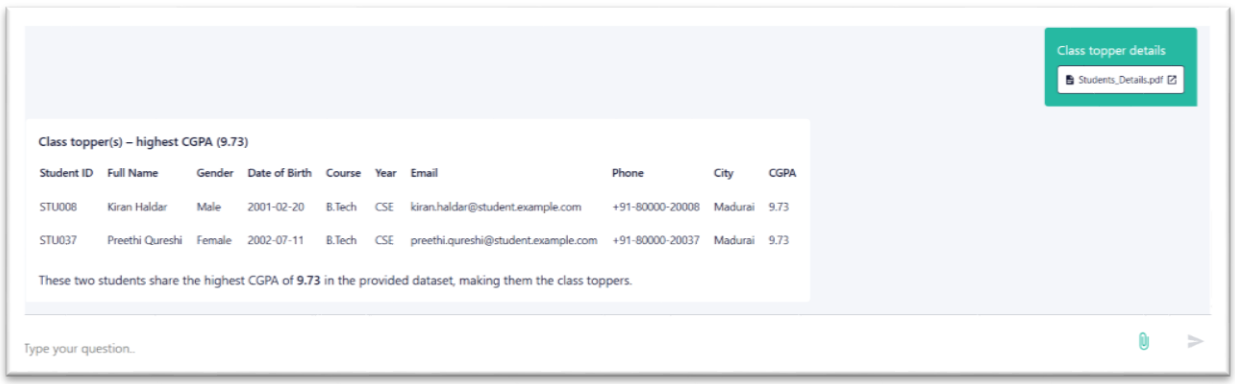

Personal / Document_QA_WF
Publish
Editor
Executions
Evaluations
Extract PDF Content
Store Document by Session
Prepare Document and Question
Receive User Input
Check Document Upload
Generate Document-...
Retrieve Document by Session
LLM – Groq
Hide chat
Chat
Session: 0d033...
Salary (INR): 1,500,000
Location: Delhi
Above average students details
Students_Details.pdf
Type message, or press 'up' for previous one
Logs
Clear execution
LLM – Groq Success in 7.458s 6,399 Tokens
Input
Output
INPUT
OUTPUT
Average CGPA (all 50 students): 7.95
Students whose CGPA is above the average (CGPA > 7.95) are listed below.
Student ID
Full Name
Gender
Date of Birth
Course
Year
Email
STU001
Arjun Ahuja
Male
2002-05-01
B.Tech CSE
2
arjun.ahuja@student.e
STU003
Vignesh Naidu
Male
2003-09-25
MBA
4
vignesh.naidu@studer


Personal / Document_QA_WF
Publish
Editor
Executions
Evaluations
Extract PDF Content
Store Document by Session
Prepare Document and Question
Receive User Input
Check Document Upload
Generate Document-...
Retrieve Document by Session
LLM – Groq
Hide chat
Chat
Session: 0d033...
M.Maha Vijay employee details
The requested information is not available in the uploaded document.
Type message, or press 'up' for previous one
Logs
Clear execution
LLM – Groq Success in 836ms 3,409 Tokens
Input
Output
INPUT
OUTPUT
The requested information is not available in the uploaded document.

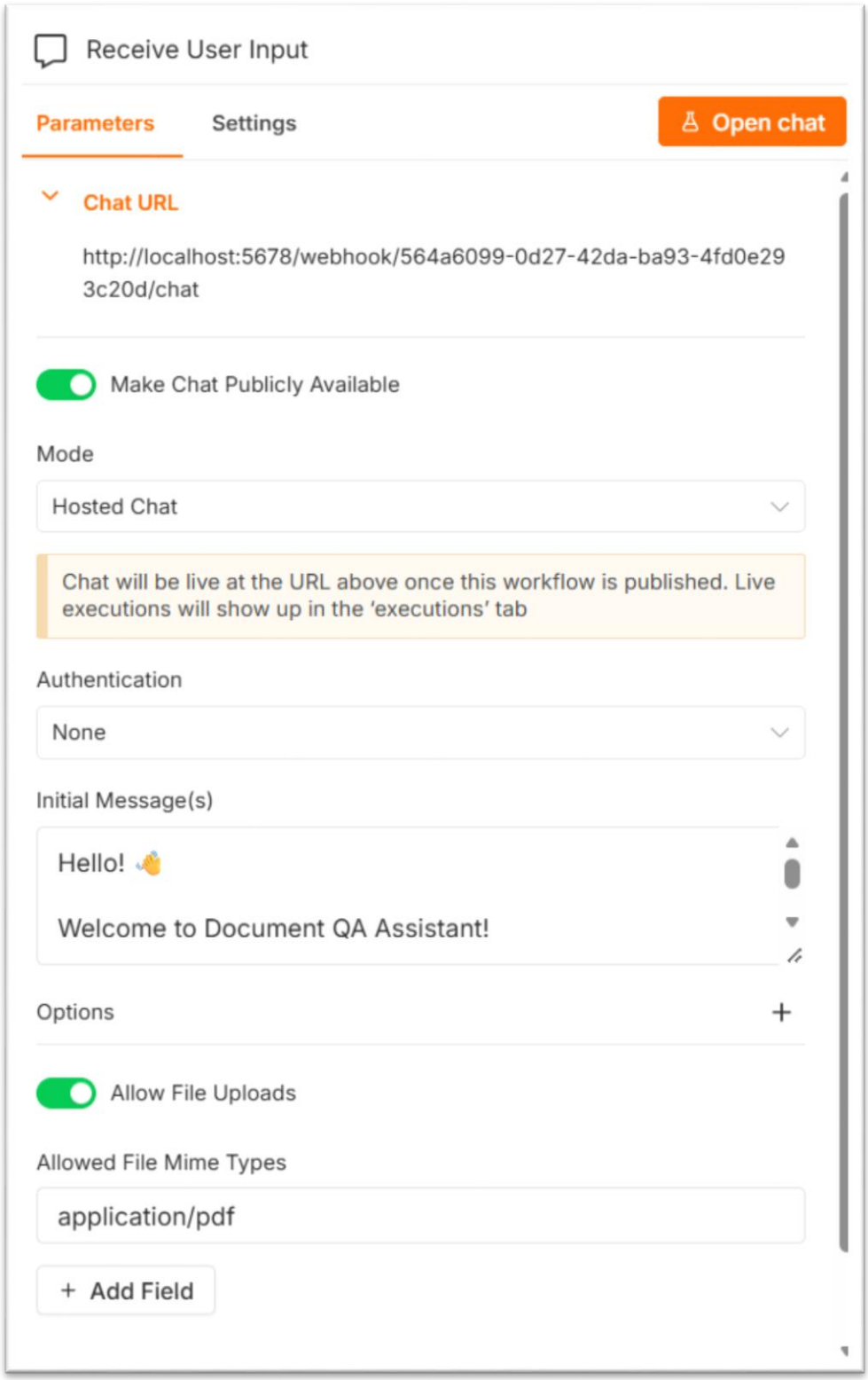

Hi there! 👋
Start a chat. We're here to help you 24/7.
Chennai employee details
Employee_Details.pdf
Chennai employee details (as listed in the uploaded document):
Emp ID Full Name Department Designation Email Phone Join Date Salary (INR) Location
EMP001 Arjun Kumar Engineering Software Engineer arjun.kumar@example.com +91-90000-10001 2019-03-15 850,000 Chennai
EMP002 Priya Sharma Engineering Senior QA Engineer priya.sharma@example.com +91-90000-10002 2018-07-01 1,100,000 Chennai
EMP004 Sneha Iyer Human Resources HR Manager sneha.iyer@example.com +91-90000-10004 2016-11-10 1,400,000 Chennai
EMP009 Suresh Babu Operations Operations Lead suresh.babu@example.com +91-90000-10009 2015-04-12 1,350,000 Chennai
EMP012 Lakshmi Narayan Human Resources Recruiter lakshmi.narayan@example.com +91-90000-10012 2023-01-09 600,000 Chennai
EMP015 Manoj Krishnan Engineering Engineering Manager manoj.krishnan@example.com +91-90000-10015 2013-02-27 2,600,000 Chennai
EMP018 Harini Subramanian Engineering QA Engineer harini.subramanian@example.com +91-90000-10018 2023-04-17 760,000 Chennai
These are all the employee records in the document that have Location: Chennai.
Type your question..

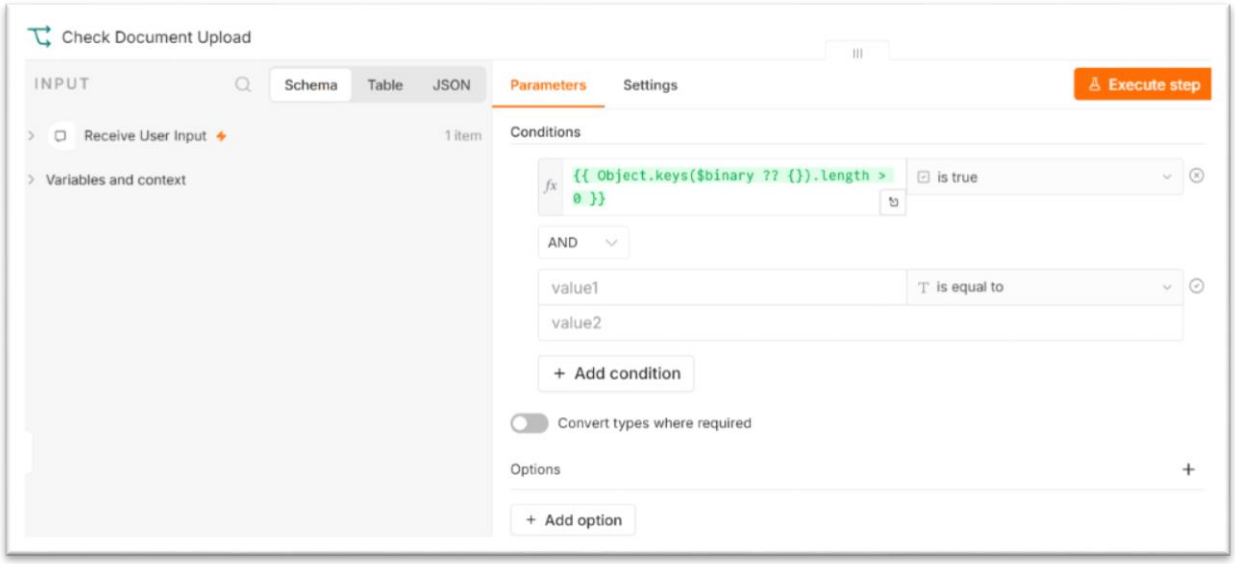

Hi there! 👋
Start a chat. We're here to help you 24/7.
These are all the employee records in the document that have Location: Chennai.
Nisha Patel's details
Nisha Patel's details
Emp ID: EMP016
Full Name: Nisha Patel
Department: Operations
Designation: Process Analyst
Email: nisha.patel@example.com
Phone: +91-90000-10016
Join Date: 2021-12-01
Salary (INR): 700,000
Location: Ahmedabad
Type your question..

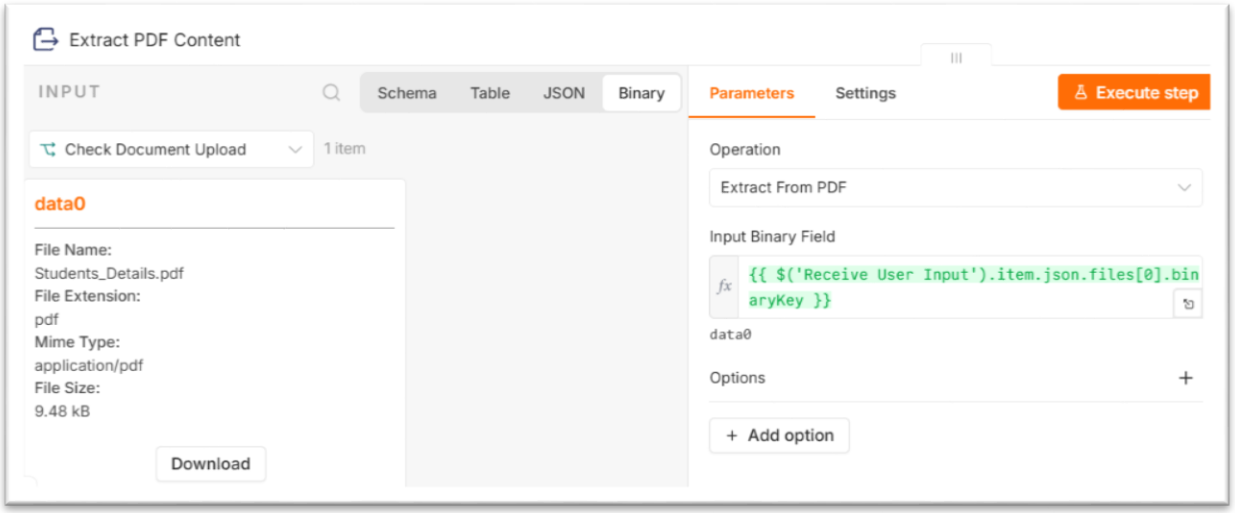

Mithun's details
The requested information is not available in the uploaded document.
ype your question..

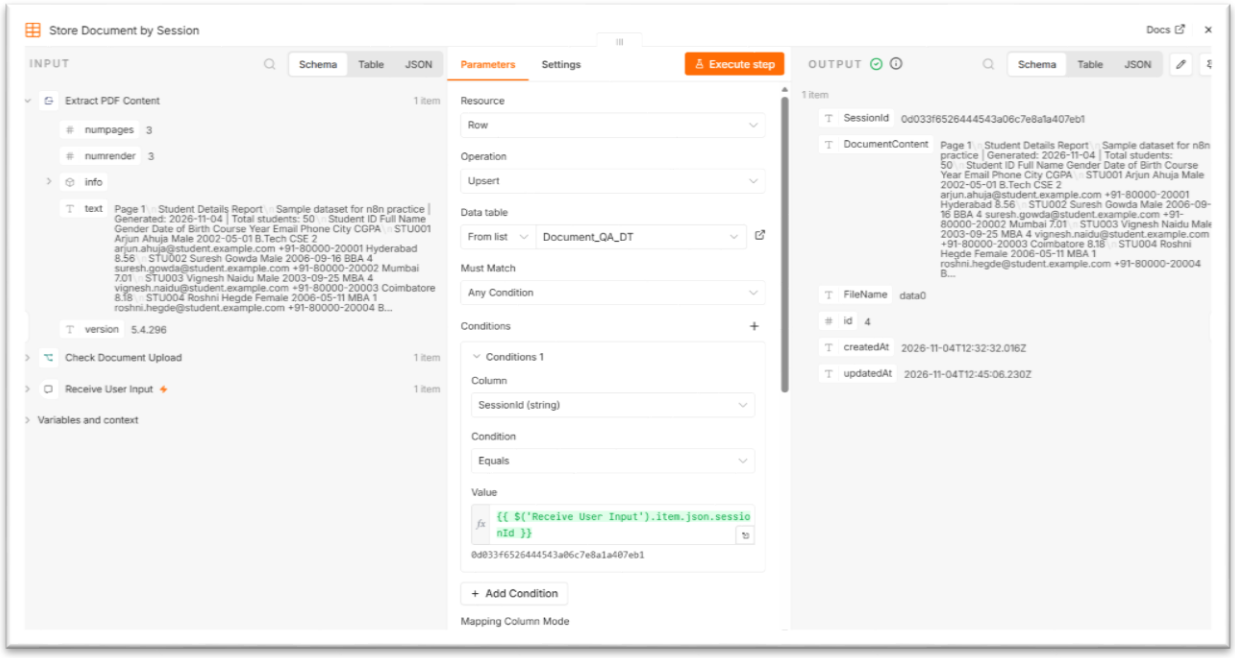

Class topper details
Students_Details.pdf
Class topper(s) – highest CGPA (9.73)
Student ID Full Name Gender Date of Birth Course Year Email Phone City CGPA
STU008 Kiran Haldar Male 2001-02-20 B.Tech CSE kiran.haldar@student.example.com +91-80000-20008 Madurai 9.73
STU037 Preethi Qureshi Female 2002-07-11 B.Tech CSE preethi.qureshi@student.example.com +91-80000-20037 Madurai 9.73
These two students share the highest CGPA of 9.73 in the provided dataset, making them the class toppers.
Type your question..

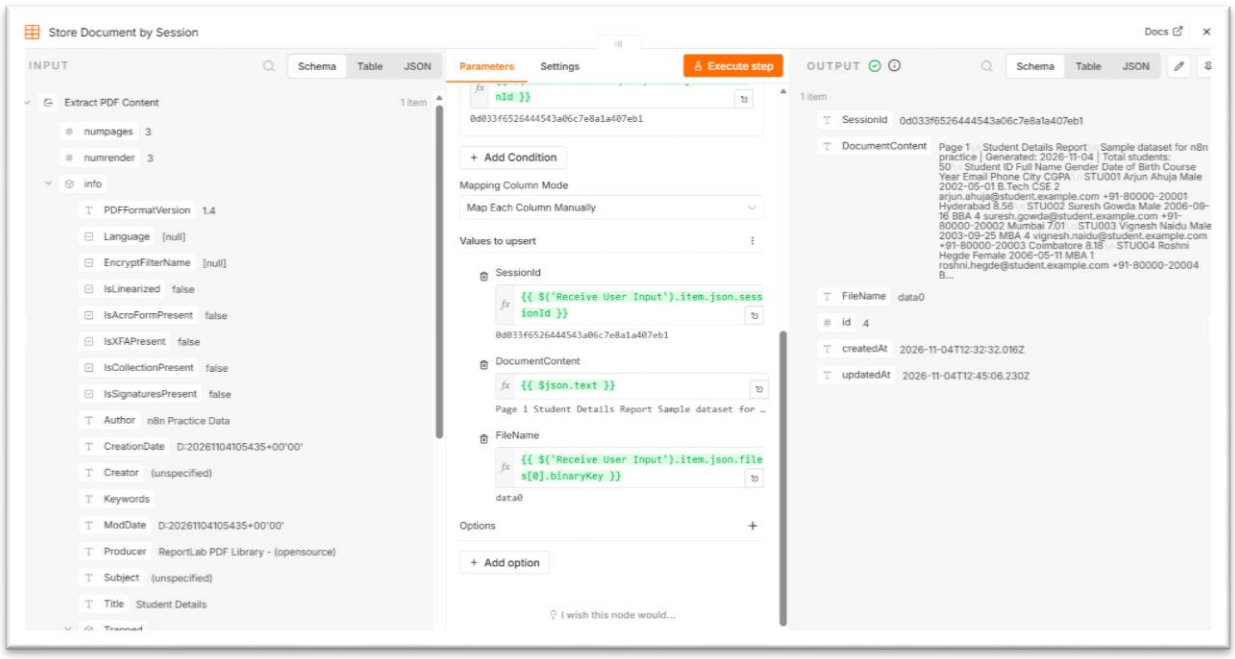
Node Parameters:

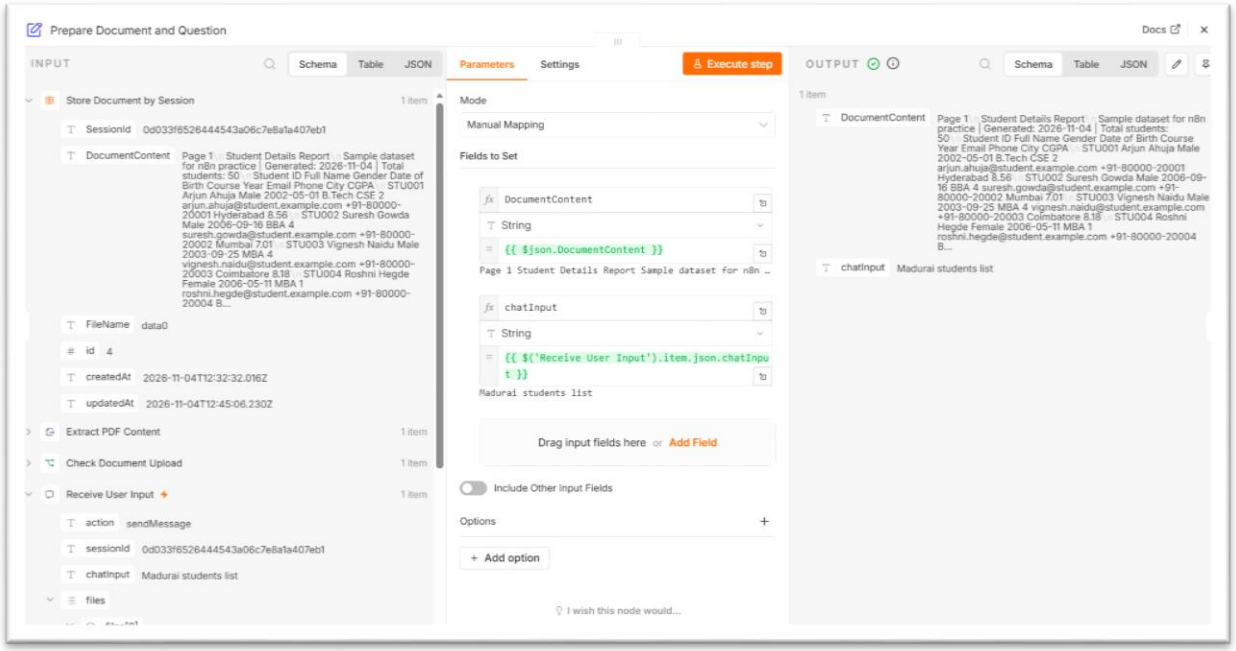


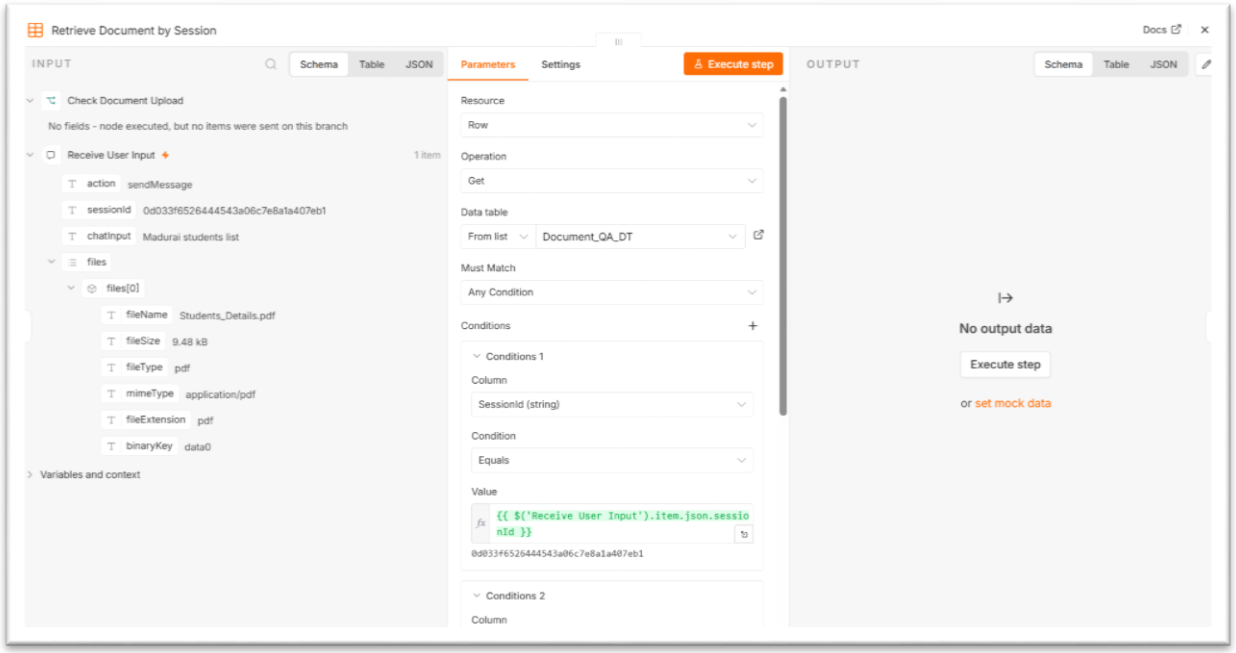

Check Document Upload
INPUT
Schema
Table
JSON
Receive User Input
1 item
Variables and context
Parameters
Settings
Execute step
Conditions
{{ Object.keys($binary ?? {}).length > 0 }}
is true
AND
value1
is equal to
value2
+ Add condition
Convert types where required
Options
+ Add option

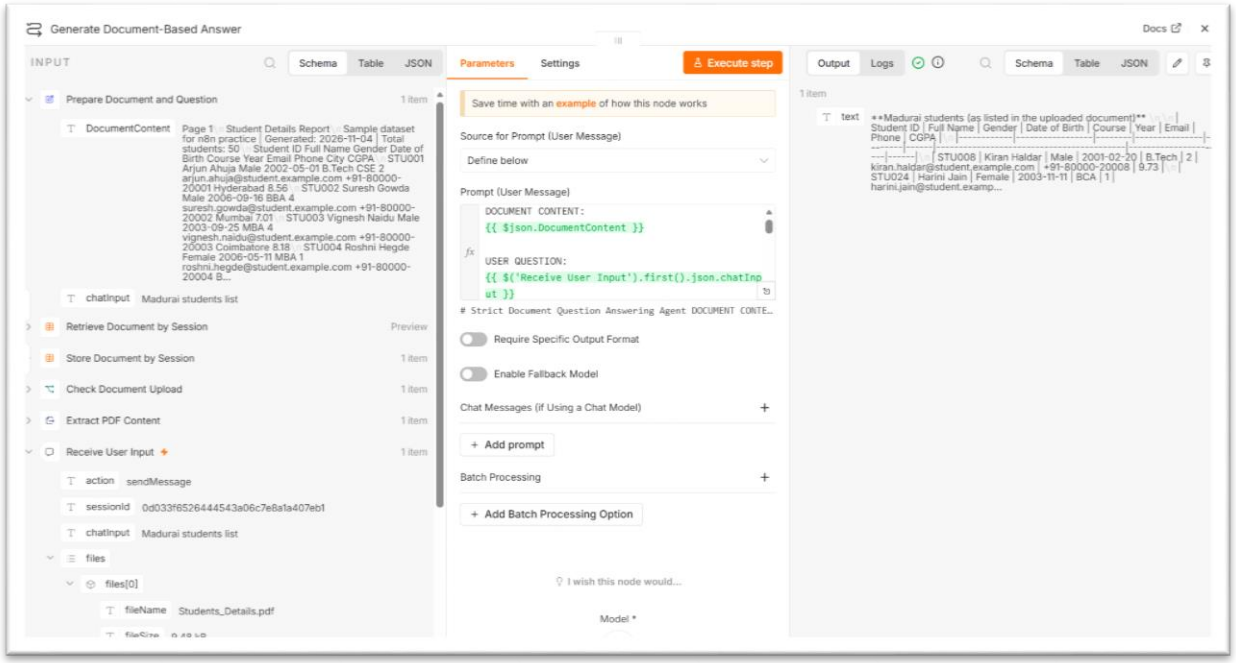

Extract PDF Content
INPUT
Schema
Table
JSON
Binary
Check Document Upload
1 item
data0
File Name:
Students_Details.pdf
File Extension:
pdf
Mime Type:
application/pdf
File Size:
9.48 kB
Download
Parameters
Settings
Execute step
Operation
Extract From PDF
Input Binary Field
{{ $('Receive User Input').item.json.files[0].binaryKey }}
data0
Options
+ Add option


Store Document by Session
Docs
INPUT
Schema
Table
JSON
Parameters
Settings
Execute step
OUTPUT
Extract PDF Content
1 item
numpages 3
numrender 3
info
text Page 1 Student Details Report Sample dataset for n8n practice | Generated: 2026-11-04 | Total students: 50 Student ID Full Name Gender Date of Birth Course Year Email Phone City CGPA STU001 Arjun Ahuja Male 2002-05-01 B.Tech CSE 2 arjun.ahuja@student.example.com +91-80000-20001 Hyderabad 8.56 STU002 Suresh Gowda Male 2006-09-16 BBA 4 suresh.gowda@student.example.com +91-80000-20002 Mumbai 7.01 STU003 Vignesh Naidu Male 2003-09-25 MBA 4 vignesh.naidu@student.example.com +91-80000-20003 Coimbatore 8.18 STU004 Roshni Hegde Female 2006-05-11 MBA 1 roshni.hegde@student.example.com +91-80000-20004 B...
version 5.4.296
Check Document Upload
1 item
Receive User Input
1 item
Variables and context
Resource
Row
Operation
Upsert
Data table
From list
Document_QA_DT
Must Match
Any Condition
Conditions
Conditions 1
Column
SessionId (string)
Condition
Equals
Value
{{ $('Receive User Input').item.json.sessionId }}
0d033f6526444543a06c7e8a1a407eb1
Add Condition
Mapping Column Mode
1 item
SessionId 0d033f6526444543a06c7e8a1a407eb1
DocumentContent Page 1 Student Details Report Sample dataset for n8n practice | Generated: 2026-11-04 | Total students: 50 Student ID Full Name Gender Date of Birth Course Year Email Phone City CGPA STU001 Arjun Ahuja Male 2002-05-01 B.Tech CSE 2 arjun.ahuja@student.example.com +91-80000-20001 Hyderabad 8.56 STU002 Suresh Gowda Male 2006-09-16 BBA 4 suresh.gowda@student.example.com +91-80000-20002 Mumbai 7.01 STU003 Vignesh Naidu Male 2003-09-25 MBA 4 vignesh.naidu@student.example.com +91-80000-20003 Coimbatore 8.18 STU004 Roshni Hegde Female 2006-05-11 MBA 1 roshni.hegde@student.example.com +91-80000-20004 B...
FileName data0
id 4
createdAt 2026-11-04T12:32:32.016Z
updatedAt 2026-11-04T12:45:06.230Z


Store Document by Session
Docs
INPUT
Schema
Table
JSON
Parameters
Settings
Execute step
OUTPUT
Extract PDF Content
1 item
numpages 3
numrender 3
info
PDFFormatVersion 1.4
Language [null]
EncryptFilterName [null]
IsLinearized false
IsAcroFormPresent false
IsXFAPresent false
IsCollectionPresent false
IsSignaturesPresent false
Author n8n Practice Data
CreationDate D:20261104105435+00'00'
Creator (unspecified)
Keywords
ModDate D:20261104105435+00'00'
Producer ReportLab PDF Library - (opensource)
Subject (unspecified)
Title Student Details
Trapped
nId }}
0d033f6526444543a06c7e8a1a407eb1
Add Condition
Mapping Column Mode
Map Each Column Manually
Values to upsert
SessionId
{{ $('Receive User Input').item.json.sessionId }}
0d033f6526444543a06c7e8a1a407eb1
DocumentContent
{{ $json.text }}
Page 1 Student Details Report Sample dataset for ...
FileName
{{ $('Receive User Input').item.json.files[0].binaryKey }}
data0
Options
Add option
I wish this node would...
1 item
SessionId 0d033f6526444543a06c7e8a1a407eb1
DocumentContent Page 1 Student Details Report Sample dataset for n8n practice | Generated: 2026-11-04 | Total students: 50 Student ID Full Name Gender Date of Birth Course Year Email Phone City CGPA STU001 Arjun Ahuja Male 2002-05-01 B.Tech CSE 2 arjun.ahuja@student.example.com +91-80000-20001 Hyderabad 8.56 STU002 Suresh Gowda Male 2006-09-16 BBA 4 suresh.gowda@student.example.com +91-80000-20002 Mumbai 7.01 STU003 Vignesh Naidu Male 2003-09-25 MBA 4 vignesh.naidu@student.example.com +91-80000-20003 Coimbatore 8.18 STU004 Roshni Hegde Female 2006-05-11 MBA 1 roshni.hegde@student.example.com +91-80000-20004 B...
FileName data0
id 4
createdAt 2026-11-04T12:32:32.016Z
updatedAt 2026-11-04T12:45:06.230Z


Prepare Document and Question
Docs
INPUT
Schema
Table
JSON
Store Document by Session
1 item
SessionId
0d033f6526444543a06c7e8a1a407eb1
DocumentContent
Page 1 Student Details Report Sample dataset for n8n practice | Generated: 2026-11-04 | Total students: 50 Student ID Full Name Gender Date of Birth Course Year Email Phone City CGPA STU001 Arjun Ahuja Male 2002-05-01 B.Tech CSE 2 arjun.ahuja@student.example.com +91-80000-20001 Hyderabad 8.56 STU002 Suresh Gowda Male 2006-09-16 BBA 4 suresh.gowda@student.example.com +91-80000-20002 Mumbai 7.01 STU003 Vignesh Naidu Male 2003-09-25 MBA 4 vignesh.naidu@student.example.com +91-80000-20003 Coimbatore 8.18 STU004 Roshni Hegde Female 2006-05-11 MBA 1 roshni.hegde@student.example.com +91-80000-20004 B...
FileName
data0
id
4
createdAt
2026-11-04T12:32:32.016Z
updatedAt
2026-11-04T12:45:06.230Z
Extract PDF Content
1 item
Check Document Upload
1 item
Receive User Input
1 item
action
sendMessage
sessionId
0d033f6526444543a06c7e8a1a407eb1
chatInput
Madurai students list
files
Parameters
Settings
Execute step
Mode
Manual Mapping
Fields to Set
DocumentContent
String
{{ $json.DocumentContent }}
Page 1 Student Details Report Sample dataset for n8n …
chatInput
String
{{ $('Receive User Input').item.json.chatInput }}
Madurai students list
Drag input fields here or Add Field
Include Other Input Fields
Options
+ Add option
I wish this node would...
OUTPUT
Schema
Table
JSON
1 item
DocumentContent
Page 1 Student Details Report Sample dataset for n8n practice | Generated: 2026-11-04 | Total students: 50 Student ID Full Name Gender Date of Birth Course Year Email Phone City CGPA STU001 Arjun Ahuja Male 2002-05-01 B.Tech CSE 2 arjun.ahuja@student.example.com +91-80000-20001 Hyderabad 8.56 STU002 Suresh Gowda Male 2006-09-16 BBA 4 suresh.gowda@student.example.com +91-80000-20002 Mumbai 7.01 STU003 Vignesh Naidu Male 2003-09-25 MBA 4 vignesh.naidu@student.example.com +91-80000-20003 Coimbatore 8.18 STU004 Roshni Hegde Female 2006-05-11 MBA 1 roshni.hegde@student.example.com +91-80000-20004 B...
chatInput
Madurai students list


Retrieve Document by Session
Docs
INPUT
Schema
Table
JSON
Check Document Upload
No fields - node executed, but no items were sent on this branch
Receive User Input
1 item
action
sendMessage
sessionId
0d033f6526444543a06c7e8a1a407eb1
chatInput
Madurai students list
files
files[0]
fileName
Students_Details.pdf
fileSize
9.48 kB
fileType
pdf
mimeType
application/pdf
fileExtension
pdf
binaryKey
data0
Variables and context
Parameters
Settings
Execute step
Resource
Row
Operation
Get
Data table
From list
Document_QA_DT
Must Match
Any Condition
Conditions
Conditions 1
Column
SessionId (string)
Condition
Equals
Value
{{ $('Receive User Input').item.json.sessionId }}
0d033f6526444543a06c7e8a1a407eb1
Conditions 2
Column
OUTPUT
Schema
Table
JSON
No output data
Execute step
or set mock data


Generate Document-Based Answer
Docs
INPUT
Schema
Table
JSON
Prepare Document and Question
1 item
DocumentContent
Page 1 Student Details Report Sample dataset for n8n practice | Generated: 2026-11-04 | Total students: 50 Student ID Full Name Gender Date of Birth Course Year Email Phone City CGPA STU001 Arjun Ahuja Male 2002-05-01 B.Tech CSE 2 arjun.ahuja@student.example.com +91-80000-20001 Hyderabad 8.56 STU002 Suresh Gowda Male 2006-09-16 BBA 4 suresh.gowda@student.example.com +91-80000-20002 Mumbai 7.01 STU003 Vignesh Naidu Male 2003-09-25 MBA 4 vignesh.naidu@student.example.com +91-80000-20003 Coimbatore 8.18 STU004 Roshni Hegde Female 2006-05-11 MBA 1 roshni.hegde@student.example.com +91-80000-20004 B...
chatInput
Madurai students list
Retrieve Document by Session
Preview
Store Document by Session
1 item
Check Document Upload
1 item
Extract PDF Content
1 item
Receive User Input
1 item
action
sendMessage
sessionId
0d033f8526444543a06c7e8a1a407eb1
chatInput
Madurai students list
files
files[0]
fileName
Students_Details.pdf
Parameters
Settings
Execute step
Save time with an example of how this node works
Source for Prompt (User Message)
Define below
Prompt (User Message)
DOCUMENT CONTENT:
{{ $json.DocumentContent }}
USER QUESTION:
{{ $('Receive User Input').first().json.chatInput }}
Strict Document Question Answering Agent DOCUMENT CONTE...
Require Specific Output Format
Enable Fallback Model
Chat Messages (if Using a Chat Model)
Add prompt
Batch Processing
Add Batch Processing Option
I wish this node would...
Model *
Output
Logs
Schema
Table
JSON
1 item
text
Madurai students (as listed in the uploaded document)
Student ID | Full Name | Gender | Date of Birth | Course | Year | Email | Phone | CGPA |
STU008 | Kiran Haldar | Male | 2001-02-20 | B.Tech | 2 | kiran.haldar@student.example.com | +91-80000-20008 | 9.73 |
STU024 | Harini Jain | Female | 2003-11-11 | BCA | 1 | harini.jain@student.examp...